

On Faultization: Rubik.
What Perturbation Reveals About the Scale of Competence
in an Exactly Analyzable Collective

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Abstract

Fault tolerance asks whether a designed function survives damage. This paper develops the reversed question, under the proposed name faultization: decompose a centralized algorithm into locally competent parts, damage them deliberately, and use the damage as an instrument for locating where competence resides and at what scale a goal-directed description begins to predict. The method needs a substrate on which nothing can hide, and the pocket cube supplies one: 3,674,160 reachable states, all enumerable, with an exact distance oracle and every controller's complete dynamics computable. On this substrate the paper runs the program to completion. Seven corner agents, each knowing its own correct pose and predicting every move's effect on itself perfectly, collectively solve 61 states of 3,674,160, a basin of 0.0017 percent, while 758,286 states, 20.6 percent of the space, are local optima where no agent will consent to move; at 72 percent of states no agent is satisfied at all, and the group's parity forbids exactly 6 of 7 agents from ever being content, so the last step of success is always a 2-agent coincidence. The cheap repairs fail measurably: permitting lateral moves converts the local optima into an ecology of 145,953 attractors with cycles up to length 4 and leaves the basin at 61; one remembered action leaves it at 61; the best of 48 heterogeneous weight draws reaches 136 by tie-breaking alone. What moves the basin is damage: deleting 3 of the 7 voters raises the mean basin to 91.8 and the best deletion pattern reaches 260, quadruple the intact collective, while the same fault class contains patterns that cut the basin to 19. Action noise multiplies success 115-fold at its interior optimum of 0.4 and the product is still 0.0019 within 200 steps, against 0.000046 for the random walk with a stop rule that solves everything eventually. Deleting one face's moves strands a corner algebraically: recovery becomes impossible from 99.2 percent of states, the exact boundary between incompetence and impossibility. Populations with conflicting targets, under strict arbitration, are ascent on a mixed potential, provably cycle-free, and at every split both encoded targets survive as equilibria among roughly 750,000 compromises neither faction encodes. Retargeting the same controller to any reachable pose leaves the basin at 61, by group conjugacy: the separation of competency machinery from target is exact, and it buys nothing. Every result is deflationary, and that is the instrument working: a claim of emergent collective agency that cannot survive this microscope on 3,674,160 states has no business being made about a tissue.

The instrument

Fault tolerance asks one question of damage: did the specified service survive it. The question has half a century of good answers, and every answer treats the fault as an enemy. This paper develops the complementary question. What does the damage make visible? When a centralized algorithm is deliberately decomposed into locally competent parts, and the parts are deleted, frozen, miscalibrated, given incompatible goals, or denied information, the scientifically interesting outcome is neither survival nor failure. It is what the perturbation reveals: where the competence actually resides, what the arbitration layer was silently contributing, which attractors the dynamics harbored all along, and at what scale a goal-directed description begins to predict rather than decorate. Call the method faultization. The word is offered as a proposal; no established field carries it.

A faultization experiment is a tuple: a decomposition into agents, an observability assignment, an actuation model, a communication budget, a target representation, a fault model, and an arbitration rule. Omit any coordinate and claims of emergent competence become unidentifiable, because the competence may live in the unexamined scheduler, the aggregation rule, the target detector, or the training procedure. The method also demands a sharp vocabulary, inherited from the dependability tradition's separation of fault, error, and failure (Avizienis, Laprie, Randell and Landwehr, 2004). Noise is transient corruption of a signal. A fault is a defect capable of producing an error. Damage is a persistent structural alteration. Local incompetence is an intact policy failing its own objective. Distributed partial competence, the interesting condition, is components that each solve a real subproblem while none holds the global policy. These are 5 different interventions with different consequences, and pooling them under "errors" is how weak studies manufacture strong conclusions.

The precedent

The direct ancestor is the Levin group's self-sorting arrays, and the method owes it both the idea and the cautionary tale, so the reconstruction has to be exact (Zhang, Goldstein and Levin, 2025). Classical Bubble, Insertion, and Selection sorts were reimplemented as collections of cell agents, one per array entry, each carrying a position, a fixed value, and a fixed local policy, each running in its own thread. Faults were frozen cells, in two grades: passive cells that cannot initiate a move but can be displaced, and stuck cells that cannot move at all. The distributed implementations often retained more final order than their centralized counterparts, the ranking of algorithms reversed between the two fault grades, mixtures of policies sorted anyway and transiently clustered by policy, and populations given opposite sorting directions settled into stable macroscopic compromises. All of this is real. None of it is trivial.

The accounting is the cautionary tale. The implementation kept a main thread that activates, monitors, and kills the agents, a global lock that serializes every swap, and a coin flip in front of each lock acquisition, which together amount to a random fair scheduler and a centralized arbitration resource. Only the Bubble cells are strictly local; the Insertion cells read every cell to their left plus a global property of the cells to their right; the Selection cells carry explicit positional targets. Decentralized execution and local information are different properties, and only one was fully delivered. The most quoted statistic, a delayed-gratification measure built from temporary decreases in sortedness, rose for Bubble cells from 0.24 with no frozen cells to 0.37 with 3. The authors themselves note that a random walker also moves away from its target, call for barrier-conditioned controls, and state plainly that the implementation can be classified with random sorting algorithms and that there is no magic beyond the rules. The neutral name for the observed effect is barrier-correlated nonmonotonic progress, and nothing in the published record yet separates it from stochastic wandering under an asynchronous scheduler. That separation needs a substrate where every alternative explanation can be enumerated. Which is what a cube supplies.

What 1974 already knew

Recovery without a dedicated fault handler has a name and a date. A protocol is self-stabilizing when every execution from every initial state eventually reaches a legitimate state, and legitimate states lead only to legitimate states: convergence plus closure, formalized by Dijkstra in 1974 with locally interacting processes on a token ring (Dijkstra, 1974; Schneider, 1993). Randomization can create convergence where deterministic symmetry forbids it (Herman, 1990). Fairness does real work there. Fault containment sharpened the metric: recovery should cost in proportion to the fault, never in proportion to the system (Ghosh, Gupta, Herman and Pemmaraju, 1996). Byzantine analysis taught how to state what a collective guarantees against components that lie (Lamport, Shostak and Pease, 1982). The locality literature proved that what bounded-radius agents can compute at all depends on anonymity, identifiers, and scheduling, before any question of efficiency arises (Naor and Stockmeyer, 1995; Angluin, Aspnes, Diamadi, Fischer and Peralta, 2006). The oldest result of all says architecture can make a system more reliable than any of its parts (Von Neumann, 1956), and biology adds that the strong form of such robustness runs through degeneracy, structurally different components with overlapping functions, rather than through copies (Edelman and Gally, 2001).

So a collective that regulates toward a target without any fault detector demonstrates, by itself, a property formalized 5 decades ago. The bar for anything stronger sits higher. Faked height is common: neural cellular automata that regrow damaged patterns were explicitly trained on damaged intermediates, so their repair was implemented, just not legibly (Mordvintsev, Randazzo, Niklasson and Levin, 2020). Evolved cellular automata performing global computation are the candid older cousin (Mitchell, Crutchfield and Hraber, 1994). The two ordinary explanations for beneficial noise, annealing and stochastic resonance, complete the checklist of deflationary accounts any emergence claim must defeat (Kirkpatrick, Gelatt and Vecchi, 1983; Gammaitoni, Hänggi, Jung and Marchesoni, 1998).

The microscope

The fixed-corner pocket cube is the $2 \times 2 \times 2$ with one corner held still: 7 moving corners, a group of $7!$ times 3 to the 6th, which is 3,674,160 elements, generated by the 9 face turns of the 3 faces that avoid the fixed corner. In an absolute frame the count is 88,179,840; the quotient by the 24 rigid rotations is the honest object for a controller that should not need a compass. The full $3 \times 3 \times 3$, for scale, has 43,252,003,274,489,856,000 states and a known diameter of 20 in the half-turn metric (Rokicki, Kociemba, Davidson and Dethridge, 2014). Its optimal solvers run on pattern databases, which is what centralized control with global information looks like (Korf, 1997; Joyner, 2008). The pocket cube's virtue is that everything fits in memory. This paper's implementation reproduces the published distance shells digit for digit, 1, 9, 54, 321, 1,847, 9,992, 50,136, 227,536, 870,072, 1,887,748, 623,800, 2,644, summing to 3,674,160 with diameter 11, and that reproduction is the calibration gate for every number below (Figure 1, left).

Four small results set the rules of evidence, and each kills a tempting inference. A fixed sequence of

face turns is a permutation of the state set, so it maps distinct states to distinct states, and no open-loop word can send every configuration to solved; feedback is necessary. But the cheapest feedback suffices for the weakest success: a random walk that tests after each move whether the target is reached, and stops there, converges from every state with probability 1. Universal convergence alone is therefore nearly worthless as evidence of competence, and the control it mandates appears below with its measured price. Third, on any finite connected graph a spanning tree rooted at the target defines a feedback policy whose every step decreases tree distance, so a globally informed controller with the target as unique attractor always exists. The group forbids nothing. Whatever fails under locality fails for informational reasons. Fourth, reversibility obstructs nothing: each generator is a bijection, but a state-dependent controller applies different generators in different states, and the closed-loop map can be many-to-one, with fixed points, basins, and cycles. One bookkeeping fact completes the setup: if a face must turn at every step, the solved state cannot be absorbing, so the controller carries a no-op, and homeostasis means staying rather than merely arriving.

Local correctness does not compose

The first experiment gives the agents everything a generous designer could defend. Each of the 7 moving corners knows its own target pose, knows its current pose, and predicts perfectly whether any candidate move would place it home. Each corner votes for the moves that would. A target-blind, state-blind arbiter sums the votes and executes the first move in a fixed order that strictly increases the number of satisfied agents, and stops when nothing does. This is distributed partial competence in its purest form: every agent is locally infallible, and nobody holds the global policy.

The closed loop was computed for all 3,674,160 states. The solved basin contains 61 states. That is 0.0017 percent. There are 758,286 non-target fixed points, 20.6 percent of the space, states where every possible move strictly decreases the number of satisfied agents and the collective therefore refuses to act. The deeper diagnosis sits in the score landscape itself: at 72 percent of all states not a single agent is satisfied, so the signal the collective climbs is silent almost everywhere, and the climb, where it exists, is short. The group adds a final insult. Its permutation and orientation constraints make a state with exactly 6 of 7 corners correct illegal, so the score 6 is unoccupied at every state and unreachable as an outcome: the last step of collective success is always a 2-agent coincidence, and a controller built from single-agent consent cannot see it coming (Figure 1, right). Local correctness, complete and error-free, does not compose into global competence, and the reason is structural: every face turn moves 4 corners at once, so the game is coupled in precisely the way single-agent scores cannot express.

Permitting lateral moves makes the picture livelier without making it better. An arbiter that moves whenever no strict decrease is available converts the 758,286 dead ends into an ecology of 145,953 attractors: 80,127 fixed points holding 1,780,448 states, 43,774 cycles of length 2 holding 1,205,629, 2,512 cycles of length 3 holding 104,887, and 19,540 cycles of length 4 holding 583,196. The solved basin stays at 61. A minor arbitration choice reorganized a fifth of the space into 145,953 distinct attractors and did nothing for competence, which is the first demonstration of the method's central

point: the arbiter is part of the system, and unexamined arbitration is where conclusions go to die.

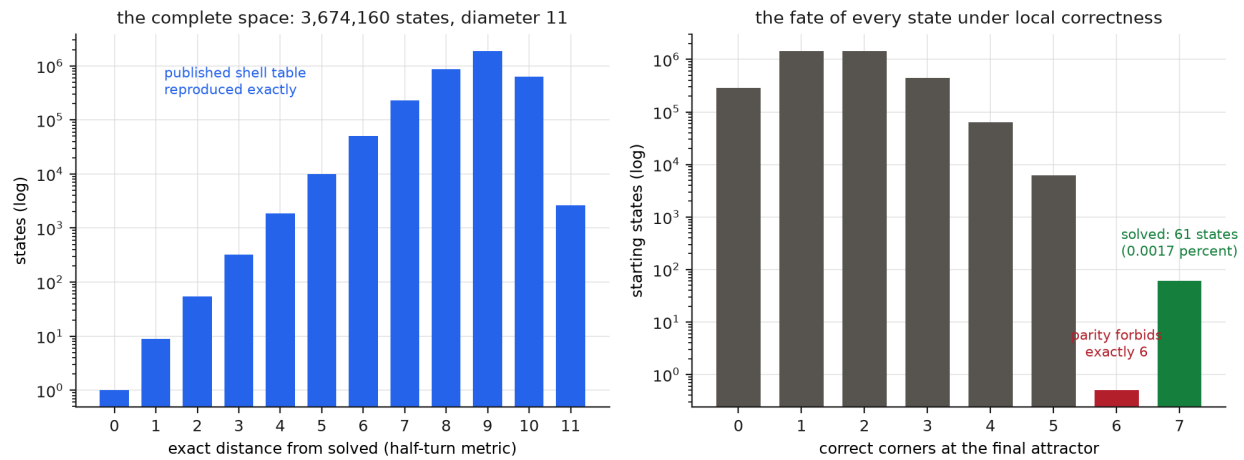


Figure 1: The microscope and the first result. Left: the complete state space by exact distance from solved; the published shell table is reproduced digit for digit, diameter 11. Right: the fate of all 3,674,160 states under strict local-correctness voting: the solved basin is 61 states, and the group's parity forbids exactly 6 of 7 corners from ever being correct.

The repairs that do not repair

Three deterministic repairs suggest themselves, and the microscope prices each. Memory first: give the collective 1 remembered action and forbid undoing it, the textbook cure for 2-cycles, run exhaustively on the augmented space of 36,741,600 state-memory pairs. The solved basin is 61. Nothing was bought. The 2-cycles dissolve and the trajectories find other traps. Heterogeneity next: draw the 7 agents' voting weights at random, 48 seeded draws. The best draw reaches a basin of 136, the worst 24, the median 64 against the homogeneous 61. Weights move the basin the way rearranging tie-breaks moves it, in both directions, with no tendency to competence. The lateral-move variant, already priced above, is the third: livelier dynamics, identical basin.

What a little randomness buys is real and bounded. Mixing the deterministic controller with a uniform random face turn at rate epsilon, computed by exact iteration over the full state vector, gives success within 200 steps of 0.000017 at epsilon 0, rising to 0.0019 at the interior optimum of epsilon 0.4, and falling to 0.000046 at epsilon 1 (Figure 2, left). The curve is the annealing curve, and the two endpoints are the controls that discipline its reading. The right endpoint is the trivial universal solver: a random walk with a stop rule, which converges from every state with probability 1 and delivers 0.000046 of that promise within a useful horizon. The interior optimum beats it 42-fold, so the voting landscape does carry real information, and beats the deterministic controller 115-fold, so exploration does real work. And the product is still 0.0019. A 115-fold improvement on microscopic is microscopic. Anyone tempted to read the interior optimum as the collective becoming intelligent has the two numbers to answer.

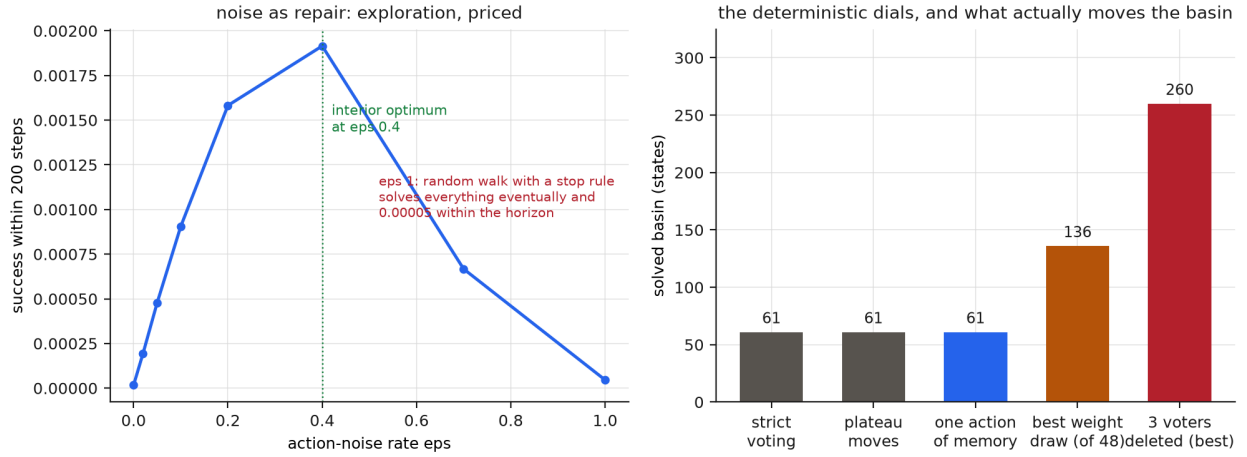


Figure 2: The repairs, priced. Left: success within 200 steps against action-noise rate; the interior optimum at 0.4 is the annealing curve's, and the epsilon 1 endpoint is the random-walk control that solves everything eventually and 0.000046 of everything now. Right: the deterministic dials all leave the basin at 61; only tie-breaking (weights) and damage (deletion) move it.

Faults as instruments

Now the faults, used as the method intends: to reveal.

Deleting voters reveals that the intact collective was over-coordinated. Remove 1 of the 7 agents and the mean basin over the 7 deletion patterns rises to 62.9; remove 2, to 69.8; remove 3, to 91.8, with the best 3-deletion pattern reaching 260, more than 4 times the intact collective, and the worst falling to 19 (Figure 2, right). Degrading component-level participation genuinely increases system-level competence here, which is the provocative effect the method went looking for, and the exhaustive analysis hands over its mechanism at once: fewer votes flatten the score landscape less finely, fewer strict-improvement demands mean fewer premature stops, and the surviving coalition's greed happens, for some patterns, to entrain toward solved. It is tie-breaking rather than insight, and the same fault class that quadruples the basin also cuts it by two thirds, depending on which agents fail. Under the noise channel the direction reverses. Success at epsilon 0.1 falls monotonically, from 0.000905 intact to 0.000333 with 3 deleted, so the redundant votes were doing real work for the explorer even as they suffocated the deterministic climber. One fault class, two opposite verdicts, decided by whether the other repair channel is on. That is the instrument at work.

Deleting moves instead of voters reveals the boundary between incompetence and impossibility. Remove one face's 3 generators and the survivors generate a subgroup whose orbit through solved contains 29,160 states, 0.79 percent of the space: from the other 99.2 percent, no controller of any intelligence can recover, because the fixed corner's neighbor on the deleted face can no longer be moved at all. Remove 2 faces and the orbit is 4 states. This is an algebraic fact, computed as the orbit of the surviving generators. It separates what no behavioral experiment can: failure that indicts the controller from failure that indicts the physics (Figure 3, left). A faultization study that

cannot make this separation will attribute to stupidity what group theory has already decided.

Conflicting targets reveal what arbitration does to normative conflict. Give m agents the solved target and 7 minus m a target B at distance 6 , and keep the strict arbiter. The summed vote is then a mixed potential, the dynamics are ascent on it, and cycles are provably impossible, which the exhaustive classification confirms at every split. What the enumeration adds is the ecology's scale and shape: roughly $730,000$ to $780,000$ equilibria at every split, nearly all of them compromises that neither faction encodes, with the count largest, $777,582$, at the most balanced split of 4 against 3 , where each faction's basin shrinks to about as small as the other's, 55 against 53 . Both encoded targets remain equilibria at every ratio; conflict is settled by basins, never by extinction, and balance fractures the landscape further instead of resolving it (Figure 3, right). Soften the arbiter to permit lateral moves and $128,244$ cycles up to length 4 reappear: oscillating truces the strict potential forbade. Every one of these outcomes is a recurrent class of competing update rules, predicted in kind, though not in count, by the potential structure; nothing requires a negotiating collective, and the enumeration is exactly what shows that nothing does.

Retargeting reveals a factorization, and its limit. The controller is fully target-relative: hand every agent a new pose from any reachable configuration and the machinery is untouched. Group translation then makes the new closed loop conjugate to the old one, so the basin of the new target equals the basin of the old, and the computation confirms it: 61 for target B , 61 for target C , 61 for solved. The separation of competency mechanism from target specification, the property Levin's framework prizes in morphogenesis, holds here in full, provably, and by construction (Levin, 2022; Fields and Levin, 2022). And it buys no competence whatsoever. A mechanism can be flawlessly reprogrammable and uniformly weak, which means reprogrammability and competence are independent axes, and evidence for one is no evidence for the other.

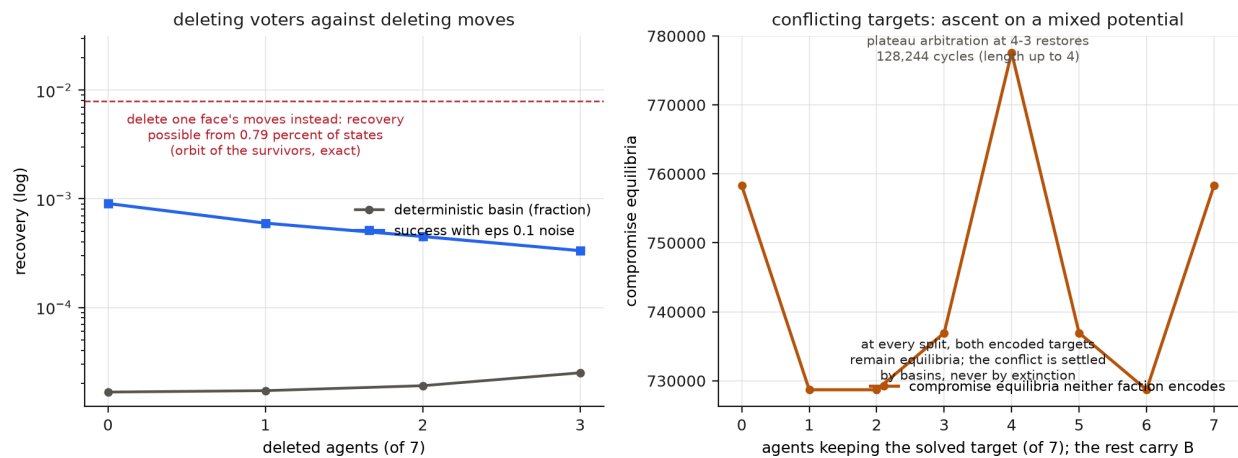


Figure 3: Faults as instruments. Left: deleting voters raises the deterministic basin while lowering noisy success, and deleting one face's moves makes recovery algebraically impossible from 99.2 percent of states, the exact boundary between incompetence and impossibility. Right: conflicting targets under strict arbitration: compromise equilibria by split, with both encoded targets surviving at every ratio.

The ladder

Claims about goal-directedness come in grades, and the cube should be placed honestly. Call G0 optimization: actions statistically improve an objective. G1 stabilization: broad convergence and closure, Dijkstra's property. G2 robust regulation: recovery from perturbation and component faults, homeostasis in the cybernetic sense that begins with purpose as feedback (Rosenblueth, Wiener and Bigelow, 1943) and runs through the regulator theorem, every good regulator being a model of what it regulates (Ashby, 1956; Conant and Ashby, 1970). G3 flexible control: multiple routes, context-sensitive action, useful temporary regression. G4 reprogrammable competence: the same machinery redirected across targets and unseen faults without redesign. G5 autonomous normativity: a system that constitutes and maintains its own conditions of existence, the enactive bar of individuality, normativity, and asymmetry that no software cube clears (Barandiaran, Di Paolo and Rohde, 2009). The controllers above sit at G0, reach G1 only on 0.0017 percent of the space, and demonstrate the G4 factorization in a form that proves the axis independent of competence. Between the rungs runs the pragmatic rule: goal language is licensed where it predicts, which is the intentional stance (Dennett, 1987), and the middle vocabulary of control mechanisms often does the work with less metaphysics (Bich and Bechtel, 2022).

What would count as an emergent collective competency, here or anywhere? The strong condition asks more than that no individual can solve the task, which is easy to arrange. It requires that no component contains the global policy, the arbiter does not either, the interaction is causally necessary, the collective generalizes to targets and faults absent from its construction, and a macro-level goal model predicts intervention outcomes more compactly than the flat micro-description. What would not count, and the cube generates every item in exact form: a random walk with an absorbing target; a distance oracle translated into votes; a target-aware arbiter; performance improvements from ordinary annealing; a stable limit cycle described in the vocabulary of preference; robustness that is duplicate votes; success dependent on a benevolent scheduler. The strong condition is demanding, and nothing in this paper meets it. That is the finding, and it is the instrument working as designed: the exhaustive substrate produces the complete deflationary catalog that any stronger claim, made on any substrate, must defeat item by item.

Is faultization a method

The traditions the term must exceed are real and close: fault injection validates recovery machinery (Hsueh, Tsai and Iyer, 1997), chaos engineering stresses deployed services (Basiri et al., 2016), lesion and knockout studies map function to component, and self-stabilization owns recovery-without-a-handler outright. The term keeps its place under 7 conditions, distilled from the failures the cube made visible. Scale must be an experimental variable, the decomposition varied rather than assumed. Fault classes must be causally separated: the paper's deletion, structural, and normative faults gave 3 different sciences. The arbiter must be audited, since a single relaxation of arbitration reorganized a fifth of the state space. Matched search controls must be run, since the random walk with a stop rule converges universally and the annealing curve explains every beneficial noise level. Novelty must be held out. Macro-level goal description must pay in compression and

prediction. And the faults must be read as instruments: the question is what structure becomes inferable because of the fault, as when deletion's two opposite verdicts located the redundancy, or the generator loss drew the exact line between the controller's failure and the group's refusal.

Under those conditions the method is a real addition, and its statement is one sentence. Ordinary fault tolerance asks whether a specified service persists; faultization asks what the fault reveals about the causal and descriptive scale of the service, including whether a different service, a different attractor, or a different agent has quietly appeared.

Objections

The strongest objection is that every result here was foreseeable: greedy local voting fails on a coupled combinatorial group, noise anneals, deletion perturbs tie-breaking, potentials cannot cycle. Foreseeable is a hindsight word. The seed intuitions this paper tested predicted that memory would enlarge the basin and that noise would rescue deleted collectives, and both predictions were wrong in the measured direction; the deletion effect ran opposite to the noise effect, which nobody's foresight supplied; and the exact sizes, 61, 758,286, 145,953, 29,160, are facts no qualitative argument delivers. The second objection is that the substrate is unfair to emergence: rigid cubies, fixed topology, no growth, no metabolism, an externally specified target. Correct, and stated as a scope condition rather than conceded as a flaw: the cube is a microscope for the deflationary catalog, and a phenomenon that survives it earns the right to be tested on richer substrates, where the same accounting will be harder and more necessary. The third objection is that the paper's deflationary tone prejudges Levin's program. It does the opposite. The program's own best methodological instincts, barrier-conditioned controls, matched stochastic baselines, the authors' plain statement that there is no magic beyond the rules, are here executed to completion on a substrate where completion is possible, and the result is a sharpened bar rather than a verdict about tissues. The fourth objection is that the strict arbiter is a straw man since real collectives are asynchronous and noisy. The plateau, memory, noise, and heterogeneity variants are precisely that objection, run exhaustively, and the basin moved from 61 only when the collective was damaged or randomized, never when it was enriched.

Falsification

The paper's claims are exhaustively computed, so falsification bears on the framework rather than the numbers. The method's usefulness fails if its 7 conditions add nothing beyond existing fault-injection practice, which would show as studies satisfying them reaching the same conclusions as studies that do not. The deflationary catalog fails as a bar if a system clears it trivially, a collective with no component holding the global policy, an audited arbiter, held-out generalization, and a predictive macro-model, arising without design effort, in which case the strong condition was too weak and needs raising. The substrate claim fails if the pocket cube's lessons do not transfer: if controllers tuned to clear this microscope are helpless on the 3x3x3 or on any substrate with growth and changing topology, the microscope measured its own glass. And the specific measured surprises invite direct attack: a memory design that does enlarge the basin, a repair channel under

which deletion helps both the climber and the explorer, or a chimera arbitration under which an encoded target goes extinct would each overturn a finding this paper presents as a lesson.

The microscope's verdict

The pocket cube will not become an organism, and this paper has not tried to make it one. Its 3,674,160 states did something more useful. They held still while every claim was counted. Local correctness, granted in its most flawless form, composed into almost nothing, and the group's parity explained why the final coincidence cannot be locally seen. The repairs that intuition trusts, memory, laterality, diversity, bought identically nothing, and the one that works is ordinary exploration, priced against the walk that solves everything eventually and almost nothing in time. Damage was the only intervention that improved the intact collective, for reasons the enumeration hands over without residue, and damage in another register drew the one line behavior can never draw alone, between what a controller failed to do and what a group made impossible. A perfectly reprogrammable mechanism carried its 61 states to every target it was pointed at. These are the facts a formal microscope yields, and their use is the bar they set. Wherever robust, flexible, reprogrammable goal-directedness is claimed of a collective, in arrays, in automata, in tissues, there is now a complete, enumerable catalog of the cheaper explanations, computed to the last state, that the claim must first defeat.

Reproducibility

The simulation, its 3 figures, and every model number quoted above regenerate from one command in the paper's repository (`simulation/run_all.py`). The first run builds the full transition tables in a few minutes of pure python; every experiment is then exhaustive numpy over all 3,674,160 states, deterministic throughout except the seeded weight draws, bit for bit on rerun. 27 invariants, among them the exact reproduction of the published distance shells, the parity hole at 6 correct corners, the equality of the strict, plateau, and memory basins at 61, the interior noise optimum against both endpoint controls, the monotone rise of the deterministic deletion basins against the monotone fall of the noisy ones, the exact orbit of the surviving generators, the cycle-free chimera under strict arbitration, and the exact conjugacy of retargeted basins, fail the run loudly if broken. The Levin-experiment statistics quoted in the reconstruction enter `results.json` as cited records with their source.

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